

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2023

Course: B. Tech.

Semester :VII

Branch : Electronics and Communication Engineering/Electronics and Telecommunication Engineering

Subject Code & Name: BTETPE702D and Fiber Optic Communication

Max Marks: 60

Date:04-01-24

Duration: 3 Hr.

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) A typical relative refractive index difference for an optical fiber designed for longdistance transmission is 1%. Estimate the NA and the solid acceptance angle in air for the fiber when the core index is 1.46. Further, calculate the critical angle at the core–cladding interface within the fiber. It may be assumed that the concepts of geometric optics hold for the fiber.	CO1/CO4	6
B) Explain the skew ray propagation in an optical fiber. Derive the acceptance conditions for skew rays.	CO1	6
C) List the applications of optical fiber communication.	CO1	6
Q.2 Solve Any Two of the following.		12
A) A multimode step index fiber with a core diameter of 80μm and a relative index difference of 1.5% is operating at a wavelength of 0.85μm. If the core refractive index is 1.48, estimate: (a) the normalized frequency for the fiber; (b) the number of guided modes.	CO2/CO4	6
B) Explain the different types of optical fibers with a neat diagram.	CO2	6
C) Define dispersion. Explain in detail with a schematic diagram	CO2/CO4	6
Q.3 Solve Any Two of the following.		12
A) With a schematic diagram explain the semiconductor injection LASER.	CO3	6
B) Derive the expression for LED power and efficiency.	CO3/CO4	6
C) Explain the operation of PIN diode.	CO4	6
Q.4 Solve Any Two of the following.		12
A) Explain Wavelength division multiplexing	CO4/CO5	6

B)	Explain the EDFA amplifier.	CO5	6
C)	Explain Raman amplifier.	CO5	6
Q. 5 Solve Any Two of the following.			12
A)	Explain the concept of self-phase modulation.	CO5	6
B)	Explain group velocity dispersion.	CO4/CO5	6
C)	Explain the nonlinear effects in fiber optic links	CO5	6
*** End ***			